Presentation <2>
Link to presentation PDF and code:

https://github.com/arvindkrishna87/Team4_TheAnalysts/tree/main/Presentation2

What did each of the team members do?

Note that multiple team members can work together on the same task. In that case, you may write the same task for all of them.

Hortense *(Consultant):* Researched whether UNI2's input size matters for our variable-sized epithelial patches. Explained how UNI2 turns an image of any size into one embedding: the image is cut into 14×14-pixel squares, each square becomes a list of numbers with a position tag, a transformer lets the squares compare with each other, and the CLS token collects the result as a fixed 1,536-number embedding. A bigger image gives more squares, not a bigger embedding. Compared KimiaNet and UNI2 on architecture (CNN that shrinks images 32× vs. transformer), input size (1000×1000 vs. 224×224), output size (4,096 vs. 1,536 numbers) and training data (TCGA slides vs. 200M+ tiles from 350k+ slides). Calculated how many 14×14 squares UNI2 would see at native size: 32 px gives 2×2, 64 px gives 4×4, 128 px gives 9×9, our typical 199 px patch gives 14×14, and 224 px gives 16×16. Reviewed published results (Chen et al., 2024): on a breast cancer subtyping task, UNI lost 6.3% when images shrank from 1,344 px to 224 px, compared with 18.8% for CTransPath and 32.5% for REMEDIS. Recorded the authors' warning that resizing to 224 px changes the effective magnification, and the limits of the evidence: every image tested was at least 224 px, no small patch was enlarged, and UNI2-h itself was not tested. Recommended keeping each patch's real size (UNI2's dynamic-size mode) instead of forcing 224×224, raising the minimum patch size from 32 px to 128 px (which keeps 76.7% of patches, using Zaharaa's audit), and confirming our slides' magnification, since UNI was pretrained at ×20.

Zaharaa *(Consultant):*Checked how our code handles patch size. precompute_pooled_features.py already feeds patches at real size with no resizing (lines 212–219), but the minimum is still 32 px (line 32), and UNI2 is not set up yet (only placeholder paths in precompute_embeddings.sbatch). Calculated how much UNI2's default resize to 224 px would stretch our patches: 32 px patches 7×, 64 px 3.5×, our typical 199 px patch only 1.1×. Using my Presentation 1 audit, 23.3% of our 81,627 patches would be stretched 1.75× or more. Designed a test comparing resized vs. real-size patches.

Dina *(Analyst):* Investigated why certain MIL training splits avoided severe overfitting, focusing on Team 2's Split 5, which achieved a validation loss of 0.023 at epoch 15 despite near-zero training loss. Analyzed the original .npz split files, training code, and case labels to compare benign/high-grade distributions, finding nearly identical class proportions across all five splits. Reconstructed patch and slice counts from PNG filenames using the model's original constraints (per_slice_cap=500, max_slices_per_stain=5), finding that Split 5 had the highest average capped training patch count (821.3 per case), but no clear relationship with validation performance. Examined trainer.py and saved checkpoints, confirming that validation metrics were aggregated rather than saved per case and identifying the epoch-15 best.pth checkpoint for future case-level analysis.

***Maya** (Analyst):* Developed a method to visually locate extracted patches on their original tissue slices. Created two Python scripts: generate_patch_coordinates.py, which adapts the original patch-placement algorithm to calculate and save patch coordinates in a CSV file, and visualize_slice_patch_index.py, which uses those coordinates to outline selected patches on the original tissue slice.

***Haowen Tan** (Analyst)*: Improved the model evaluation pipeline by handling missing embeddings, tracking patches by filename to verify correct attention indexing, and generating patch-level attention visualizations.

How does it help the project?

***Hortense**:* Explains how UNI2 handles images of different sizes, so the team knows what to expect when its embeddings replace KimiaNet's for our variable-sized patches. Shows that very small patches would be stretched up to 7× if resized to 224 px (using Zaharaa's audit), and gives a concrete setup to avoid that: keep the real patch size and raise the minimum patch size to 128 px.

***Zaharaa**:* Gives the team a clear rule for adding UNI2: keep the current no-resize setup and don't copy the Resize(224) step from UNI2's model card. Shows that raising the minimum to 128 px cuts the worst-case stretch from 7× to 1.75× while keeping 76.7% of patches.

***Maya** (Analyst):* This method allows us to identify where extracted patches are located on the original tissue slices, helping us better interpret model results. It also provides a way to investigate whether high-attention patches share common tissue characteristics or visual patterns.

***Haowen Tan** (Analyst)*: These improvements allow the model to complete evaluations without failing due to missing embeddings. Filename-based tracking helps ensure attention scores are correctly matched to their original image patches. The visualizations also improve model interpretability by identifying which histopathology regions contribute most to predictions.

Issues faced that haven't been resolved:

***Hortense**:* No published study tests patches as small as ours or UNI2-h specifically, so the effect of input size on our results is not confirmed. The recommendations (keep real size, raise the minimum to 128 px) have not been tested by retraining; the planned UNI2 vs. KimiaNet experiment is what would settle them. I have not confirmed how the code handles patch sizes that are not a multiple of 14 pixels, so the square counts above assume the leftover pixels are dropped. The magnification (microns per pixel) of our slides is also not yet confirmed, and it determines how much resizing to 224 px changes the scale of the tissue.

***Zaharaa**:* The test can't run until UNI2 is loaded in the code, so the effect of resizing on our results is unconfirmed. Real size isn't proven better either, since a 128 px patch gives UNI2 only

9×9 squares instead of 16×16. The slides' magnification isn't recorded in the code.

***Maya** (Analyst):* Some full-resolution TIFF files cannot be decoded by the code process (e.g., case 36), preventing visualization of those slices. While patch coordinates can still be generated using the annotated PNG files, the TIFF decoding issue prevents the corresponding full tissue slices from being visualized.

Haowen Tan** (Analyst):** The main issue is missing embeddings for some image patches. While smaller patches are expected to lack embeddings, we still need to investigate whether other missing embeddings are caused by data processing or extraction issues. These gaps may affect patch coverage and the completeness of our attention analysis.

Proposed next steps

***Hortense**:* Run the planned UNI2 vs. KimiaNet comparison twice, once with patches resized to 224 px and once at their real size, using the same seed and split, then compare accuracy and recall per class. Confirm the magnification (microns per pixel) of our slides, since UNI was pretrained at ×20. Check how the code handles patch sizes that are not a multiple of 14 pixels.

***Zaharaa**:* Raise MIN_PATCH_SIZE to 128 and keep the no-resize transform when UNI2 is added. Once UNI2 runs, embed about 50 patches per size group both ways and check whether the small ones change. If they do, train both ways and keep the option with better accuracy and recall per class. Confirm the slides' scan magnification.

***Maya** (Analyst):* Visualize high-attention patches to identify potential patterns in the regions the model focuses on. Additionally, expand the visualizations across additional cases and stains (H&E, Melan-A, and Sox10).

Haowen Tan** (Analyst):** Further investigate missing embeddings to identify whether they result from patch size limitations or other preprocessing issues. Analyze how missing embeddings affect model predictions and attention distributions. Compare attention patterns across correctly and incorrectly classified cases to identify potential weaknesses in the model and opportunities for improvement.

References

Whose work did you build up on?

***Hortense**:* Built on Zaharaa's patch-size audit of all 81,627 patches (23.3% under 128 px, typical patch about 199 px) and on the team's existing patching and feature-extraction pipeline (precompute_pooled_features.py, MIN_PATCH_SIZE on line 32). Sources used:

Chen, R. J., Ding, T., Lu, M. Y., Williamson, D. F. K., Jaume, G., Song, A. H., Chen, B., Zhang, A., Shao, D., Shaban, M., Williams, M., Oldenburg, L., Weishaupt, L. L., Wang, J. J., Vaidya, A., Le, L. P., Gerber, G., Sahai, S., Williams, W., & Mahmood, F. (2024). Towards a general-purpose foundation model for computational pathology. *Nature Medicine*, 30(3), 850–862. https://doi.org/10.1038/s41591-024-02857-3

Mahmood Lab. (n.d.). *UNI2-h* [Model card]. Hugging Face. https://huggingface.co/MahmoodLab/UNI2-h
Mahmood Lab. (n.d.). *UNI: Towards a general-purpose foundation model for computational pathology* [Source code repository]. GitHub. https://github.com/mahmoodlab/UNI
Riasatian, A., Babaie, M., Maleki, D., Kalra, S., Valipour, M., Hemati, S., Zaveri, M., Safarpoor, A., Shafiei, S., Afshari, M., Rasoolijaberi, M., Sikaroudi, M., Adnan, M., Shah, S., Choi, C., Damaskinos, S., Campbell, C. J. V., Diamandis, P., Pantanowitz, L., Kashani, H., Ghodsi, A., & Tizhoosh, H. R. (2021). Fine-tuning and training of DenseNet for histopathology image representation using TCGA diagnostic slides. *Medical Image Analysis*, 70, 102032.
Rea, K. (2022, October 12). *How to select the correct magnification and patch size for digital pathology projects*. Andrew Janowczyk. https://andrewjanowczyk.com/how-to-select-the-correct-magnification-and-patch-size-for-digital-pathology-projects/

Zaharaa: Built on my Presentation 1 patch-size audit, Hortense's explanation of how UNI2 processes images, and the team's feature-extraction code (precompute_pooled_features.py, precompute_embeddings.sbatch). Sources used:

Chen, R. J., Ding, T., Lu, M. Y., Williamson, D. F. K., Jaume, G., Song, A. H., Chen, B., Zhang, A., Shao, D., Shaban, M., Williams, M., Oldenburg, L., Weishaupt, L. L., Wang, J. J., Vaidya, A., Le, L. P., Gerber, G., Sahai, S., Williams, W., & Mahmood, F. (2024). Towards a general-purpose foundation model for computational pathology. Nature Medicine, 30(3), 850–862. https://doi.org/10.1038/s41591-024-02857-3
Mahmood Lab. (n.d.). UNI2-h [Model card]. Hugging Face. https://huggingface.co/MahmoodLab/UNI2-h

Haowen Tan *(Analyst)*: Built upon the team's existing hierarchical attention MIL training and evaluation pipeline, pretrained patch embeddings, and earlier attention visualization code. Extended these components to improve evaluation robustness, patch tracking, and model interpretability.